# 2020 AMC 10A Problem 11 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

## 2020 AMC 10A Problem 11

**Problem:**
What is the median of the following list of $4040$ numbers?

$$1, 2, 3, \ldots, 2020, 1^2, 2^2, 3^2, \ldots, 2020^2$$

**Answer Choices:**

A. $1974.5$
B. $1975.5$
C. $1976.5$
D. $1977.5$
E. $1978.5$

**Solution:**

We can see that

$$44^2 = 1936 \text{ which is less than } 2020$$

Therefore, there are

$$2020 - 44 = 1976 \text{ of the } 4040 \text{ numbers greater than } 2020$$

Also, there are

$$2020 + 44 = 2064 \text{ numbers that are less than or equal to } 2020$$

Since there are $44$ duplicates/extras, it will shift up our median's placement down $44$. Had the list of numbers been $1, 2, 3, \ldots, 4040$, the median of the whole set would be

$$\frac{1 + 4040}{2} = 2020.5$$

Thus, our answer is

$$2020.5 - 44 = \boxed{(C)\ 1976.5}.$$

$$\textbf{OR}$$

As we are trying to find the median of a $4040$-term set, we must find the average of the $2020$th and $2021$st terms Since $45^2 = 2025$ is slightly greater than $2020$, we know that the $44$ perfect squares $1^2$ through $44^2$ are less than $2020$, and the rest are greater. Thus, from the number $1$ to the number $2020$, there are $2020 + 44 = 2064$ terms. Since $44^2$ is $44 + 45 = 89$ less than $45^2 = 2025$ and $84$ less than $2020$, we will only need to consider the perfect square terms going down from the $2064$th term, $2020$, after going down $84$ terms.

Since the $2020$th and $2021$st terms are only $44$ and $43$ terms away from the $2064$th term, we can simply subtract $44$ from $2020$ and $43$ from $2020$ to get the two terms, which are $1976$ and $1977$.

Averaging the two, we get $\boxed{(C)\ 1976.5}$.